ASAD-44

Nucleophilic Substitution

Ref.: A Guide Book to Mechanism in Organic Chemistry

By

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- The classical displacement reaction exemplified by the conversion of an alkyl halide into an alcohol by the action of aqueous base.

$$HO^{\ominus} + R\text{-}Hal \longrightarrow HO\text{-}R + Hal^{\ominus}$$

- Kinetic measurements on reactions in which alkyl halides are attacked by a wide variety of different nucleophiles, Nu:, have revealed two types – one in which,

  Rate = k_2[RHal][Nu:] … … … (1) S_N2

  and another in which,

  Rate = k_1[RHal] … … … (2) S_N1

  i.e., the rate is independent of [Nu:].

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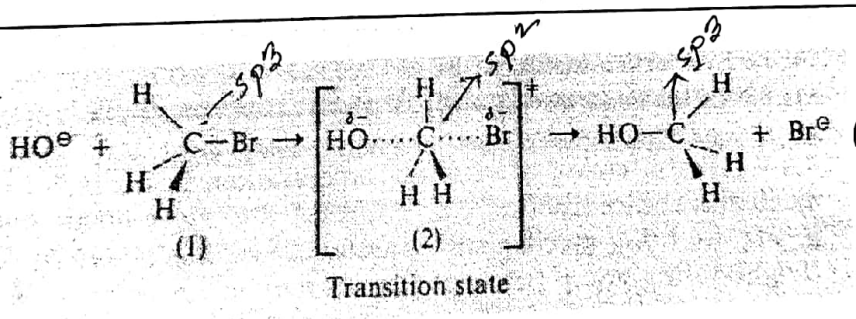
Relation of Kinetics to Mechanism

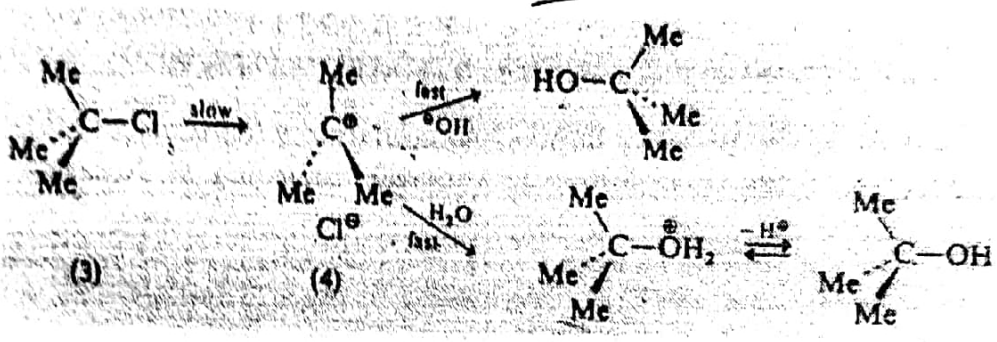
- Hydrolysis of the primary halide bromomethane in aqueous base has been shown to proceed according to equation (1). S_N2
- Ingold has suggested a transition state in which the attacking hydroxyl ion becomes partially bonded to the reacting carbon atom before the incipient bromide ion has become wholly detached from it, thus part of the energy necessary to effect the breaking of the C-Br bond is then supplied by that produced in forming HO-C bond.



- Quantum mechanical calculation shows that an approach by the hydroxyl ion along the line of centres of the carbon and bromine atoms is that of lowest energy requirement.
- The initially sp^3 hybridized carbon atom becomes sp^2 hybridized in the transition state, the HO and Br being associated with the two lobes of the unhybridized p orbital that is thereby made available.

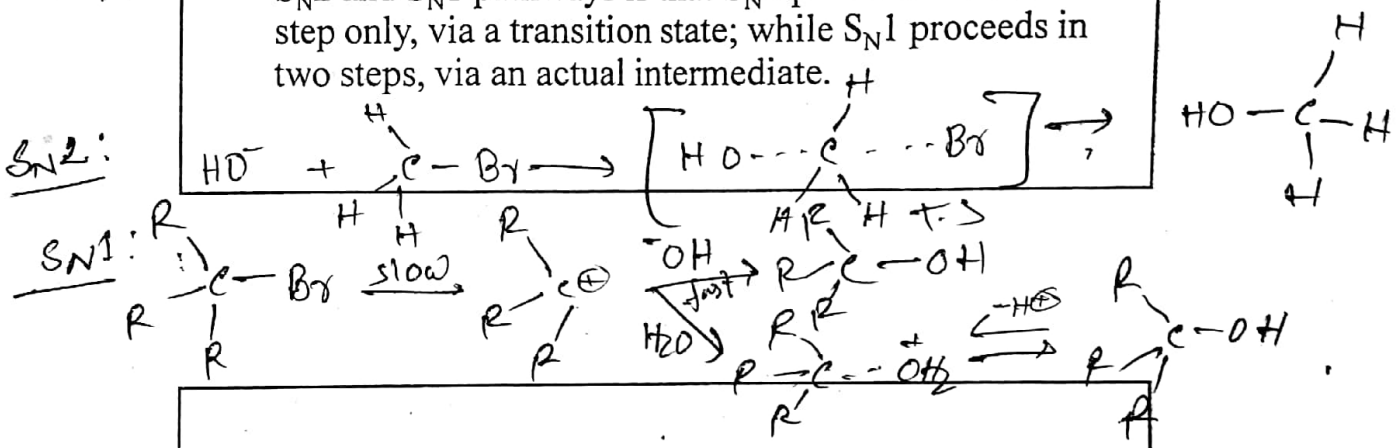
S_N1:

- This type of mechanism has been designated by Ingold as S_N2: Substitution Nucleophilic bimolecular.
- Hydrolysis of the tertiary halide, t-butyl chloride in base is found kinetically to follow equation (2), i.e., as the rate is independent of [$^-$OH].



- This has been interpreted as indicating that the halide undergoes slow ionization as the rate-limiting step to yield the ion pair R^+Cl^-; followed by rapid, non rate-limiting attack by $^-$OH or, if that is suitable, by solvent, the latter often predominating because of its very high concentration.
- This type of mechanism has been designated as S_N1: Substitution Nucleophilic unimolecular.
- The energy necessary to effect the initial ionization is largely recovered from the energy evolved through solvation of the resultant ion pair.

- The cation in the ion pair, in which the central carbon atom carries the +ve charge, is of course a carbocation intermediate, and during its formation the initially sp^3 hybridized carbon atom collapses to a more stable planar (sp^2) state, in which the three methyl groups are as far apart from each other as they can get.
- Attack by ^-OH or solvent (e.g., H_2O) can then take place from either side of this planar intermediate.
- Thus the salient difference between reaction by the S_N2 and S_N1 pathways is that S_N2 proceeds in one step only, via a transition state; while S_N1 proceeds in two steps, via an actual intermediate.



- The order is an experimentally determined quantity, the overall order of a reaction being the sum of the powers of the concentration terms that appear in the rate equation.

Rate = k_3[A][B][C]	Third order overall
Rate = k_3[A]^2[B]	Third order overall
Rate = k_2[A]^2	Second order overall

- The molecularity refers to the number of species that are undergoing bond-breaking and/or bond-making in one step of the reaction, usually in the rate-limiting step. It is important to realize that the molecularity is not an experimentally determined quantity.

- In the hydrolysis of bromomethane, order and molecularity then coincide, the reaction being second order overall and bimolecular. Order and molecularity do not always, or necessarily, have the same value.
- Simple kinetic measurements can be an inadequate guide to which of the above two mechanisms, S_N1 or S_N2 is actually operating in, for example, the hydrolysis of a halide.
- Where the solvent can act as a nucleophile (solvolysis), e.g., H_2O, we would expect for an S_N2 type reaction, Rate $= k_2[RHal][H_2O]$ but as $[H_2O]$ remains effectively constant the rate equation actually observed will be, Rate $= k_{obs}[RHal]$ and simple kinetic measurements in aqueous solution will thus suggest, erroneously, that the reaction is of the S_N1 type.

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- A kinetic distinction between the operation of the SN1 and SN2 modes can be made by observing the effect on the overall reaction rate of adding a competing nucleophile, e.g., azide anion, N_3^-.
- The total nucleophile concentration is thus increased, and for the S_N2 mode where [Nu:] appears in the rate equation, this will result in an increased reaction rate due to the increased [Nu:].
- By contrast, for the S_N1 mode [Nu:] does not appear in the rate equation, i.e., is not involved in the rate-limiting step, and adding of N_3^- will thus be without significant effect on the observed reaction rate, though it will naturally influence the composition of the product.

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S_N1 mood এর ক্ষেত্রে solvent polarity বাড়লে ion solvating ability বাড়ে ফলে rate বাড়ে।

Effect of Solvent

- Changing the solvent in which a reaction is carried out often exerts a profound effect on its rate and may, indeed, even result in a change in its mechanistic pathway.
- Thus for a halide that undergoes hydrolysis by the S_N1 mode, increase in the polarity of the solvent (i.e., increase in ϵ, the dielectric constant) and/or its ion-solvating ability is found to result in a very marked increase in reaction rate.
- Thus the rate of solvolysis of the tertiary halide, Me_3CBr, is found to be 3×10^4 times faster in 50% aqueous ethanol than in ethanol alone.

* F-16 Role of solvent in N.S TU-11
f-16,15

S_N2 mood এ solvent polarity বাড়লে নিউ ক্লিয়াস solvated অবস্থায় থাকে ফলে reaction centre এ হয়তো attack করতে সমস্যা হয় ফলে rate হ্রাস পায়।

solvent এর polarity বাড়লে নিউক্লিয়াস (Nu) এর stability বাড়ে

$$R{-}Hal \rightarrow \left[\overset{\delta+}{R}\cdots\overset{\delta-}{Hal}\right]^{\ddagger} \rightarrow R^{\oplus}Hal^{\ominus}$$

- The energy required to effect such a process decreases as ϵ rises; the process is also facilitated by increasing solvation and consequent stabilization of the developing ion pair compared with the starting material.
- That such effects, particularly solvation, are of prime importance is borne out by the fact that S_N1 type reactions are extremely uncommon in the gas phase.
- For the S_N2 mode, increasing solvent polarity is found to have a much less marked effect, resulting in a slight decrease in reaction rate.

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Nucleophile solvented থাকলে

S_N2 এ Nucleophile attacking এর সময় solvent এর polarity বাড়লে সেটা তার Nucleophile কে attack করতে বাধা দেয়। ফলে reaction rate কমে যায়।

polar hydroxylic solvent থেকে polar non hydroxylic solvent এ transfer করলে S_N2 mood এ reaction rate অনেক বাড়ে; কারণ solvent ঘিরে না weakly isolated করে মূল nu কে attack করতে পারে।

22-Oct-16

This occurs because solvation of the T.S. is likely to be somewhat less effective than that of the initial nucleophile – hence the slight decrease.

A very marked effect on the rate of S_N2 reactions is effected on transferring them from polar hydroxylic solvents to polar non-hydroxylic solvents.

Thus the reaction rate of the primary halide, MeI, with N_3^- at 0° increased 4.5×10^4 fold on transfer from methanol (ϵ = 33) to dimethyl formamide (DMF, $HCONMe_2$), with very much the same polarity (ϵ = 37).

The attacking nucleophile, N_3^-, is highly solvated through hydrogen bonding in MeOH whereas it is very much less strongly solvated and not by hydrogen bonding in $HCONMe_2$.

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The large unsolvated N_3^- anion (in $HCONMe_2$) is a very much more powerful nucleophile than when surrounded (as in MeOH) by a very much less nucleophilic solvation envelope, hence the rise in reaction rate.

Rate increases of as much as 10^9 fold have been observed on transferring S_N2 reactions from, e.g., MeOH to another polar non-protic solvent, dimethyl sulphoxide (DMSO), Me_2SO (ϵ = 46).

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S_N2 reaction এ MeOH থেকে DMSO solvent এ স্থানান্তর করলে rate 10^9 times বাড়ে।

Intermediate ion stability কম হলে S_N2 থেকে S_N1 হয়।

- So far as actual changes of mechanistic pathway with change of solvent concerned, increase in solvent polarity and ion-solvating ability may change the reaction mode $S_N2 \rightarrow S_N1$.
- Transfer from hydroxylic to polar, non-protic solvents (e.g., DMSO) can, and often do, change the reaction mode from $S_N1 \rightarrow S_N2$ by enormously increasing the effectiveness of the nucleophile in the system.

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Solvent polarity বাড়লে S_N2 mode S_N1 এ চলে যায়।

Hydroxylic polar solvent থেকে Non hydroxylic polar solvent এ transfer করলে S_N1 mode S_N2 এ চলে আসে।

Effect of Structure

CH_3-Br	$MeCH_2-Br$	Me_2CH-Br	Me_3C-Br
	1°	2°	3°

- The first and last members are described in the literature as undergoing ready hydrolysis, the two intermediate members being more resistant.
- Measurement of their rates of hydrolysis with dilute, aqueous ethanolic sodium hydroxide solution gives the plot and further kinetic investigation reveals a change in order of reaction, and hence presumable, of mechanism, as the series is traversed.

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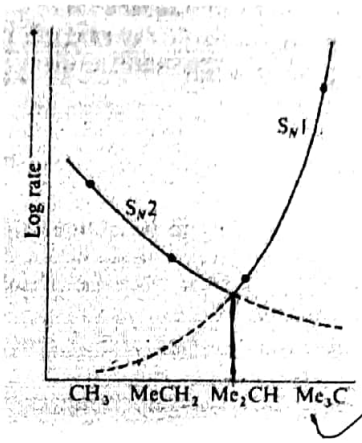
- **Thus bromomethane and bromoethane are found to follow a second order rate equation, 2-bromopropane a mixed second and first order equation, the relative proportion of the two depending on the initial [$^-$OH] (the higher the initial concentration the greater the second order proportion) and the total rate here being a minimum for the series, while 2-bromo-2-methylpropane is found to follow a first order rate equation.**



stability অনুযায়ী (electronic factor)

- We need to consider the effect on the transition state of both electronic and steric factors.

→ এখানে তিনটি কারণের জন্য সুবিধা মতো nucleophile attack করতে সমস্যা হয় (steric effect)

me
Me – C –
me

যেখানে me group e⁻ release করে সেখানে e⁻ এর positive charge এর সমতা No⁻ attack করার জন্য হয়।

electronic effect

- For S_N2 attack, the enhanced inductive effect of an increasing number of methyl groups might be expected to make the carbon atom that bears the bromine progressively less positively polarized and hence less readily attacked by $^-$OH.
- This effect is probably small and steric factors are of more importance, thus $^-$OH will find it progressively more difficult to attack the bromine carrying carbon as the latter becomes more heavily substituted.
- More significantly, the resultant S_N2 transition state will have five groups around this carbon atom compared with only four in the initial state, there will thus be an increase in crowding on going from the initial halide to the transition state, and this relative crowding will increase as the size of the original substituents increases (H → Me).

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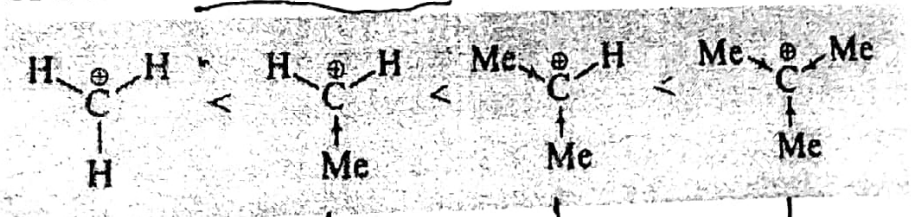
- The more crowded the T.S. relative to the starting materials, the higher its energy will be and the slower will it be formed.
  - We would thus expect the purely S_N2 reaction rate to decrease as the above series is traversed.
- It is in fact possible to effect nucleophilic substitution (Br^- + R-Cl) on a series of halides analogous to those in Figure, under conditions such that a strictly second order rate equation (S_N2 pathway) is followed throughout. We then observe:

Relative S_N2 rate:	CH_3Cl	$MeCH_2Cl$	Me_2CHCl	Me_3CCl
	1	2.7×10^{-2}	4.9×10^{-4}	2.2×10^{-5}

- For S_N1 attack, considerable charge separation has taken place in the T.S. and the ion pair intermediate to which it gives rise is therefore often taken as a model for it.
  - As the above halide series is traversed, there is increasing stabilization of the carbocation moiety of the ion pair, i.e., increasing rate of formation of the T.S.
  - This increasing stabilization arises from the operation of both an inductive effect,



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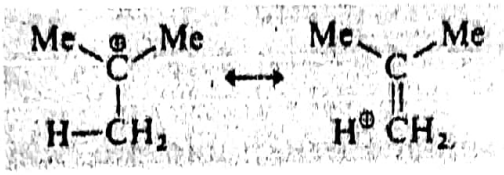
and hyperconjugation, e.g.,



via the hydrogen atoms attached to the α-carbons, the above series of carbocations having 0, 3, 6 and 9 such hydrogen atoms, respectively.

Support for such an interaction of the H-C bonds with the carbon atom carrying the positive charge is provided by substituting H by D in the original halide, the rate of formation of the ion pair is then found to be slowed down by ~10% per deuterium atom incorporated: a result compatible only with the H-C bonds being involved in the ionization.

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- This is known as a secondary kinetic isotope effect, secondary because it is a bond other than that carrying the isotope label that is being broken.
- The relative contributions of hyperconjugation and inductive effects to the stabilization of carbocations is open to debate, but it is significant that a number of carbocations will only form at all if they can take up a planar arrangement, the state in which hyperconjugation will operate most effectively.
- In steric terms there is a relief of crowding on going from the initial halide, with a tetrahedral disposition of four substituents about the sp^3 hybridized carbon atom, to the carbocation with a planar disposition of only three substituents about the sp^2 hybridized carbon atom.

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- The three substituents are as far apart from each other as they can get in the planar carbocation and the relative relief of crowding (halide → carbocation) will increase as the substituents increase in size (H → Me → Me_3C).
- The S_N1 reaction rate would thus be expected to increase markedly (on both electronic and steric grounds) as the series of halides is traversed.
- It has not proved possible to confirm this experimentally by setting up conditions such that the four halides of Figure all react via the S_N1 pathway.

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- Thus, as the S_N2 rate is expected to decrease, and the S_N1 rate to increase, across the series in Figure, the reason for the observed pattern of reaction rates, and changeover in reaction pathway, becomes apparent.
- Consider the following series:

CH_3-Cl $C_6H_5CH_2-Cl$ $(C_6H_5)_2CH-Cl$ $(C_6H_5)_3C-Cl$

- Hydrolysis in 50% aqueous acetone, a mixed second and first order rate equation is observed for phenylchloromethane (benzyl chloride) – moving over almost completely to the SN1 mode in water alone.

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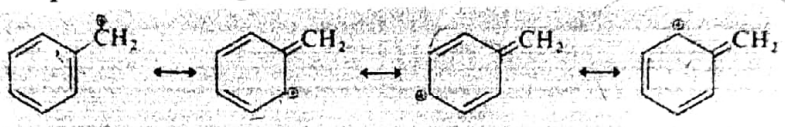
- Diphenylchloromethane is found to follow a first order rate equation with a very large increase in total rate while with triphenylchloromethane (trityl chloride) the ionization is so pronounced that the compound exhibits electrical conductivity when dissolved in liquid SO_2.
- The main reason for the greater promotion of ionization – with consequent earlier changeover to the SN1 pathway in this series – is the considerable stabilization of the carbocation, by delocalization of its positive charge.



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- This is a classical example of an ion stabilized by charge delocalization via the agency of the delocalized π orbitals of the benzene nucleus.
- The effect will become progressively more pronounced and S_N1 attack further facilitated with $(C_6H_5)_2CHCl$ and $(C_6H_5)_3CCl$ as the possibilities for delocalizing the positive charge are increased in the carbocations to which these latter halides give rise.
- S_N2 attack on the CH_2 in $C_6H_5CH_2$-Cl is found to proceed at very much the same rate as on that in $MeCH_2Cl$, suggesting that any adverse steric crowding in the T.S. by the bulky C_6H_5 group is compensated by a small electronic effect promoting reaction.

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The steric crowding in T.S due to bulky C_6H_5 group is compensated by small electronic effect promoting the reaction rate.

That's why the S_N2 attack on the CH_2 in $C_6H_5CH_2Cl$ is found to proceed at very much the same rate as like $MeCH_2Cl$.

Q. Allyl halides are more reactive than n-propyl halides in both SN1 and SN2.

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TU-16

- Similar carbocation stabilization can also occur in the hydrolysis of allyl halides, e.g., 3-chloropropene:

$$CH_2{=}CH{-}CH_2Cl \rightarrow [CH_2{=}CH{-}\overset{\oplus}{C}H_2 \leftrightarrow \overset{\oplus}{C}H_2{-}CH{=}CH_2]\ Cl^{\ominus}$$

- S_N1 attack is thus promoted and allyl, like benzyl, halides are normally more reactive than species, e.g., $CH_3CH_2CH_2Cl$ and $C_6H_5CH_2CH_2CH_2Cl$, in which such carbocation stabilization can not take place.
- By contrast, vinyl halides such as chloroethene, $CH_2{=}CHCl$ and halogenobenzenes are very unreactive towards nucleophiles.

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SN1 & SN2 ?

Factor : 4

$CH_2 = CH \cdot CH_2Cl$ → sp³

$CH_2 = CH\,Cl$ → sp²

- The halogen atom is now bonded to an sp^2 hybridized carbon with the result that the electron pair of the C-Cl bond is drawn closer to carbon than in the bond to an sp^3 hybridized carbon.
- The C-Cl is found to be stronger and thus less easily broken than, for example, CH_3CH_2Cl and the C-Cl dipole is smaller; there is thus less tendency to ionization (S_N1) and a less positive carbon for ^-OH to attack (S_N2); the π electrons of the double bond also inhibit the close approach of an attacking nucleophile.
- The double bond would not help to stabilize either the S_N2 transition state or the carbocation involved in the S_N1 pathway.

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very important

⇒ C-Cl এর double bond এর জন্য C-Cl bond এর e⁻ negative charge এর জন্য they are close to each other.

double bond এর জন্য

Nu⁻ → $CH_2{=}CH{-}Cl$ → C-Cl এর e⁻ rich হওয়ায় Nu⁻ attack করতে পারে না।

- Very much the same considerations apply to halogenobenzenes with their sp^2 hybridized carbons and the π orbital system of the benzene nucleus, their reactions, which though often bimolecular are not in fact simply S_N2 in nature.
- The influence of steric factors on the reaction pathway is particularly observed when substitution takes place at the β-position.

CH_3-CH_2-Br	$MeCH_2-CH_2-Br$	Me_2CH-CH_2-Br	Me_3C-CH_2-Br
1·0	$2·8 \times 10^{-4}$	$3·0 \times 10^{-2}$	$4·2 \times 10^{-6}$

The figures quoted are relative rates of reactions (S_N2 throughout) with EtO^- in EtOH at 55°C.

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→ Tu-16

→ reaction rate কমার কারণ বাধা হলি সাপেক্ষে B-e ত 1

কোন compound এ কোন কোন reaction rate বেশি কমে যায় (১ ও ২ structure দেখে প্রশ্ন তৈরি করো)

- Any differences in electronic effect of the Me groups through two saturated carbon atoms would be very small and the reason for the rate differences is steric: increased difficulty of approach of EtO^- from the back of the carbon atom carrying Br and increased crowding in the resultant T.S. (T.S = Transtion state)
- The reason for the particularly large drop in rate between 1-bromo-2-methylpropane and 1-bromo-2,2-dimethylpropane (neopentyl bromide) is that the T.S. for the former, though somewhat crowded, can, by rotation about the C_α-C_β bond, adopt one conformation (I) in which the attacking EtO^- is interfered with only by H, while no such relief of crowding is open in the T.S. (II) for the latter.

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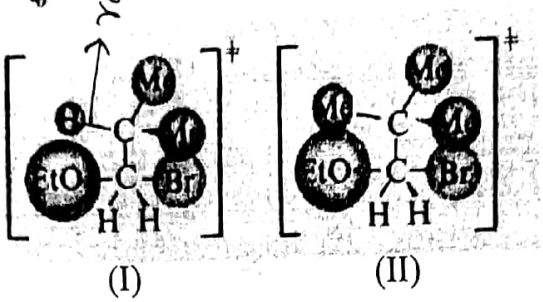
→ electronic effect কম steric effect বেশি

crowding বাড়লে Et-O- attack করা কঠিন হয়ে যায়।



(I) (II)

- The T.S. (II) will thus be at a much higher energy level, $\Delta G^{\ddagger}$ will be larger and the reaction rate correspondingly lower.
- In the S_N1 solvolysis in EtOH of 3-Chloro-1-butene, not one but a mixture of two isomeric ethers is obtained and the same mixture is also obtained from the similar solvolysis of 1-chloro-2-butene.

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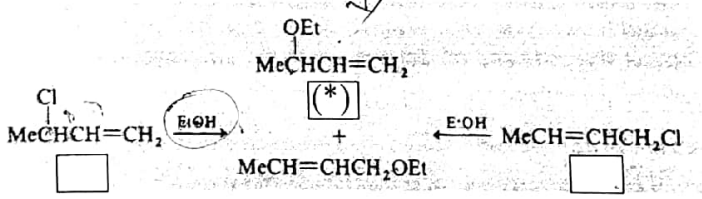

- This clearly reflects formation of the same delocalized allylic cation as an ion pair intermediate from each halide, capable of undergoing subsequent rapid nucleophilic attack by EtOH at either C_1 or C_3.

$$[MeCH^{\oplus}-CH=CH_2 \leftrightarrow MeCH=CH-CH_2^{\oplus}]\ Cl^{\ominus}$$

Allylic rearrangements have been observed in the course of displacement reactions that are proceeding by a bimolecular process. Such reactions are referred to as S_N2' and are believed to proceed:

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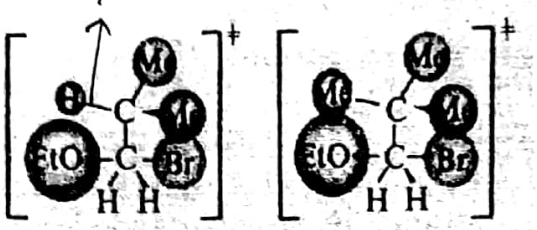
S_N2' reaction दो steps में होती है

(1) Carbocation formation

(2) Nucleophile का attack.



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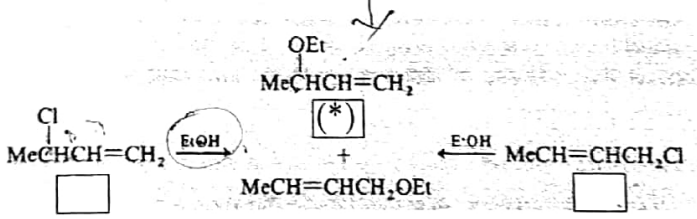

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S_N2' reaction দুই ধাপে হয়

① Carbocation formation

② Nucleophile এর attack.

∞ reaction medium ও structure এর উপর reaction নির্ভর করে।

$$Nu^{\ominus}: + CH_2=CH-\underset{R}{CH}-Cl \rightarrow Nu-CH_2-CH=\underset{R}{CH} + Cl^{\ominus}$$

- This process tends to occur when substituents (R) on the α-carbon atom are bulky enough to reduce markedly the rate of direct S_N2 displacement at C_α.
- The S_N2 hydrolysis of 1-bromo-2,2-dimethylpropane (neopentyl bromide) is slow due to steric hindrance.
- Carrying out the reaction under conditions favoring the SN1 mode can result in an increased reaction rate but the product alcohol is found to be 2-methyl-2-butanol and not the expected 2,2-dimethylpropanol (neopentyl alcohol): a neopentyl rearrangement has taken place.

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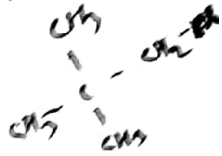
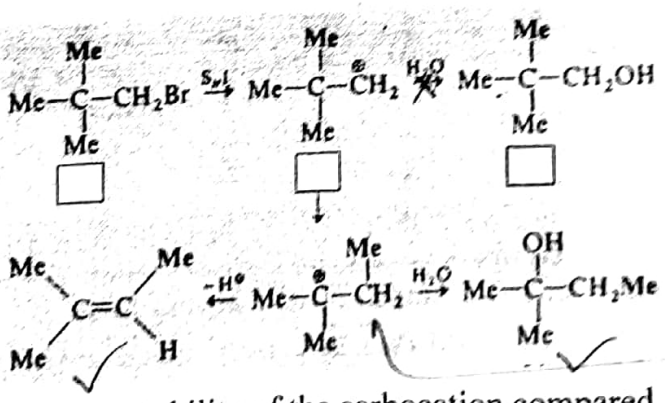
R বড় হলে reaction center এ attack করা কঠিন।



F-16

nucleophilic attack করা কঠিন → steric effect এর কারণে S_N2 না হয়ে S_N1 হয়।



- The greater stability of the carbocation compared with the initial primary one provides the driving force for the C-C bond breaking involved in migration of the Me group with its electron pair.

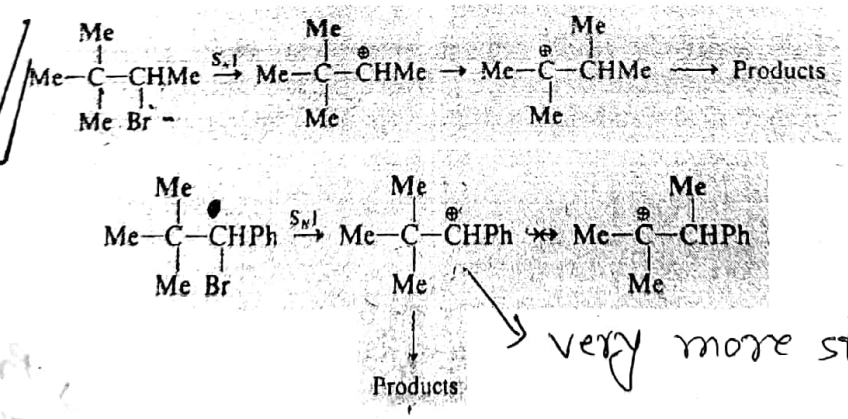
34

Primary halide stable না হওয়ার কারণ e⁻ releasing group কম, 3° halide বেশি।

- Such changes in carbon skeleton involving carbocations are known collectively as Wagner-Meerwein rearrangements.
- Further confirmation of the involvement of rearrangement is the simultaneous formation of the alkene, 2-methyl-2-butene by loss of proton: a product not obtainable without rearrangement.
- It is interesting to note that while the neopentyl type bromide undergoes rearrangement during S_N1 hydrolysis, no such rearrangement takes place with its phenyl analogue.

35

⊛ এগুলোর উদাহরণ হবে ques. থেকে মনে?

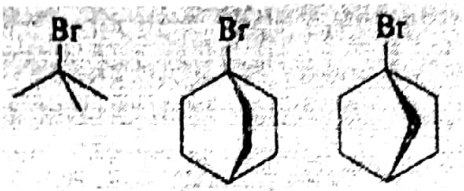


very more stable রেজোন e⁻ delocalized হয়।

- This reflects the greater stability of the benzylic cation, though only secondary, compared with the tertiary cation that would be – but in fact is not – obtained by its rearrangement.

36

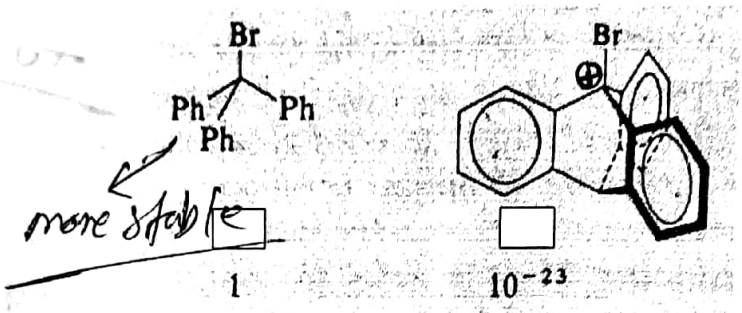
- The effect of structure on relative reactivity may be seen particularly clearly when a halogen atom is located at the bridgehead of a bicyclic system. The following rates were observed for solvolysis in 80% aqueous ethanol at 25°C.



- All the tertiary halides so that by the S_N2 mode would not be expected to occur. For S_N2, attack from back, not possible for cage like structure, transitions state also not possible because of their fairly rigid framework.

- Solvolysis via ion pair (S_N1) as happens with t-butyl bromide is also inhibited for last two structures because resultant carbocations would be very difficult to form for their rigid framework, not possible to make planar stable carbocations. These carbocations are of very much higher energy level than usual and are formed only slowly.
- In case of the following two structures, the rate of reaction is greatly reduced for second one because of the greater rigidity to form carbocations. For second structure, π-delocalization of carbocation with phenyl can not occur because the carbons are not in the same plane.

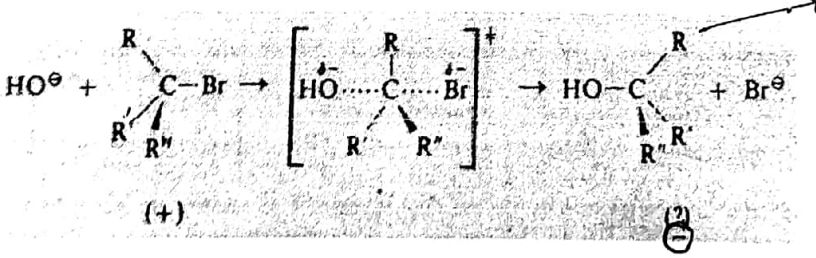
Tri-phenyl carbo cation are more stable why? then tri-methyl carbo cation ?



tertiary methyl bromide এবং tertiary phenyl bromide এর rate difference কত হবে?

39

S_N2 Mechanism: Inversion of Configuration



configuration টা change করেছে।

- The carbon atom is said to have undergone inversion of its configuration.
- If the product could be the bromide, instead of the corresponding alcohol, it would be found to rotate the plane of polarization of plane polarized light in the opposite direction, i.e., (-), to the starting material, (+), for it would, of course, be its mirror image.

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- The actual product is the alcohol and we are unfortunately not able to tell, merely by observing its direction of optical rotation, whether it has the same or the opposite configuration to the bromide from which it was derived.
- In order to confirm that the above S_N2 reaction is, in practice, attended by an inversion of configuration, as theory requires, it is necessary to have an independent method for relating the configuration of starting material and product, e.g., the bromide and corresponding alcohol above.

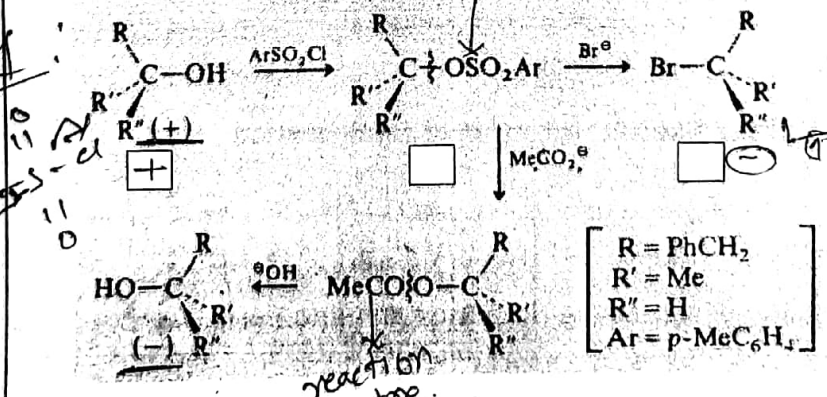
41

Q. S_N2 reaction inversion of configuration occur Explain with example ?

TU-16
F-10

Determination of Relative Configuration

Ex:



inversion of configuration

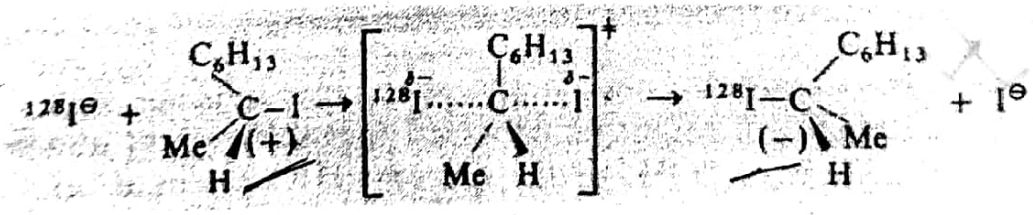
(S_N2 ला inversion configuration होते)

42

- In the series of reactions on the optically active (+) alcohol, formation of an ester with 4-methylbenzenesulphonyl (tosyl) chloride is known not to break the C-O bond of the alcohol (the case may be shown by using an alcohol labeled with ^{18}O in its OH group and demonstrating that this atom is not eliminated on forming the tosylate), hence the tosylate must have the same configuration as the original alcohol.
- Reaction of this ester (tosylate) with $MeCO_2^-$ is known to be a displacement in which $ArSO_3^-$ is expelled and $MeCO_2^-$ introduced, hence the C-O bond is broken in this reaction and inversion of configuration can thus take place in forming the acetate.

- Alkaline hydrolysis of the acetate can be shown not to involve fission of the alkyl-oxygen linkage (hydrolysis of an acetate in which the alcohol-oxygen atom is ^{18}O labeled fails to result in the latter's replacement), so the alcohol must have the same configuration as the acetate.
- As the product is found to be the mirror image of the starting material, opposite direction of optical rotation, an inversion of configuration must have taken place during the series of reactions and can have occurred only during reaction of $MeCO_2^-$ with tosylate.

- Reaction of this tosylate with a number of nucleophiles showed that inversion of configuration occurred in each case: it may thus be concluded with some confidence that it occurs on reaction with Br^- to yield the bromide, i.e., that the bromide, like the acetate, has the opposite configuration to the original alcohol.
- The bimolecular (S_N2) displacement reactions are attended by inversion of configuration – was established in an elegant and highly ingenious experiment, in which an optically active alkyl halide undergoes displacement by the same – though isotopically labeled – halide ion as nucleophile, e.g., radioactive $^{128}I^-$ on (+) 2-iodooctane.



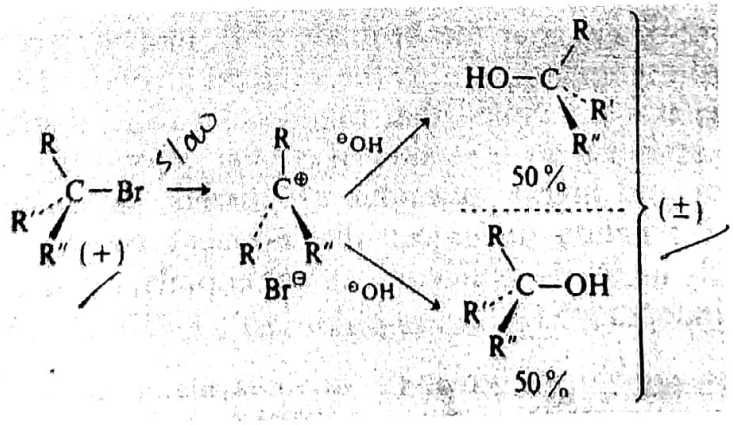
- The displacement was monitored by observing the changing of radioactive ^{128}I between the inorganic (sodium) iodide and 2-iodooctne, and it was found, under these conditions, to be second order overall.
- If inversion takes place, as S_N2 requires, the optical activity of the solution will decline to zero, i.e., racemization will occur.

100% inversion 27

- This will happen because inversion of the configuration of a molecule of (+) results in formation of a molecule of its mirror image (-), which 'pair off' with a second molecule of (+) to form a (±) racemate: the observed rate of raemization will thus be twice the rate of inversion.
- Each act of bimolecular displacement must thus proceed with inversion of configuration.

47

S$_N$1 Mechanism: Racemization



48

S_N1 mood এ racemization and some degree of inversion হয়।

22 Oct 16

As the carbocation formed in the slow, rate-limiting step of the reaction is planar, it might be expected that subsequent attack by a nucleophile such as ^-OH, or the solvent (H_2O:), would take place with equal readiness from either side of this planar carbocation; leading to a 50/50 mixture of species having the same and the opposite configuration as the starting material, i.e., that racemization would take place yielding an optically inactive (±) product.

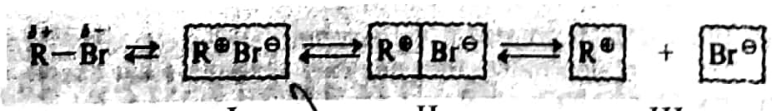
The expected racemization is rarely observed, it is almost always accompanied by some degree of inversion.

49

carbocation এ strong Recimate mixture হবে, inversion হবে।

Nucleophile এ strong Recimate mixture এ তবে inversion হবে।

- The relative proportions of the two are found to depend on: (a) the structure of the halide, in particular the relative stability of the carbocation to which it can give rise and (b) the solvent, in particular on its ability as a nucleophile.
- The more stable the carbocation, the greater is the proportion of racemization; the more nucleophilic the solvent, the greater is the proportion of inversion.
- These observations become understandable if the rate-limiting S_N1 ionization follows the sequence:



Inversion হচ্ছে।

Inversion হচ্ছে।

racemic mixture হচ্ছে।

* Br- more stronger than (e) তাই Completely Recised All inversion.

50% inversion এ 50% Recimate mixture হচ্ছে।

- Here I is an intimate ion pair in which the jointly solvated gegen-ions are in very close associated with no solvent molecules between them, II is a solvent-separated ion pair and III represents the dissociated and separately solvated pair of ions.
- In the solvolysis reaction, attack on R^+ by a solvent molecule, e.g., H_2O: in I is likely to lead to inversion, as attack can take place on the 'back' side of R^+, but not on the 'front' side where there are no solvent molecules, and which is shielded by the Br^- gegen ion.
- Attack in II is more likely to lead to attack from either side, leading to racemization, while attack on III can clearly happen with equal facility from either side.

51

- Thus the longer the life of R^+, i.e., the longer it escapes nucleophilic attack, the greater the proportion of racemization that we should expect to occur.
- The life of R^+ is likely to be longer the more stable it is – (a) above, but the shorter the more powerfully nucleophilic the solvent – (b) above.
- Thus solvolysis of (+) $C_6H_5CHMeCl$, which can form a stabilized benzyl type carbocation, leads to 98% racemization while (+) $C_6H_{13}CHMeCl$, where no comparable stabilization can occur, leads to only 34% racemization.

52

we know, the proportion of racemization depends on the stability of the carbocation. the more stable the carbocation the greater is the proportion of racemization. Here for (+) $C_6H_5CHMeCl$ a stabilized benzyl type carbocation is formed. For this reason the solvolysis of (+) $C_6H_5CHMeCl$ leads to 98% racemization.

22-Oct-16

On the other hand, in solvolysis of $C_6H_5CHMeCl$ such stabilization of carbocation do not occur and hence resulting lower proportion of recimization (34%)

we know more the nucleophilic the solvent the greater is the proportion of inversion and lower the proportion of racemization.

- Solvolysis of (+) $C_6H_5CHMeCl$ in 80% acetone/20% water leads to 98% racemization but in the more nucleophilic water alone to only 80% racemization.
- It is important to notice that racemization is clearly very much less of a stereochemical requirement for S_N1 reactions than inversion was for S_N2.

53

Since, water alone is more nucleophilic than 80% acetone, so in solvolysis of (+) $C_6H_5CHMeCl$ leads to the proportion of inversion and lower the proportion of racemization (80%)

The Mechanistic Borderline

- Secondary halides may follow a mixed first/second order rate equation.
- The question arises whether such a reaction is proceeding via both S_N2 and S_N1 pathways simultaneously (their relative proportions depending on the solvent, etc.) or whether it is proceeding via some specific, 'in between' mechanistic pathway.
- In solvolytic reactions, mixed kinetics may not be detectable, irrespective of what is actually happening, as both S_N1 and S_N2 pathways are likely to follow a rate equation of the form: Rate = k[R-X].
- In some cases it is possible to demonstrate that a 'mixed' $S_N1 + S_N2$ pathway is not operating.

54

Q. How can you detect a reaction whether SN1 or SN2 by the racemization? The greater the percentage of racemization the more it will the in SN1 pathway.

22-Oct

- Thus solvolysis of the halide, (+) $C_6H_5CHMeCl$, mentioned above, but this time in $MeCO_2H$, was found to lead to 88% racemization and 12% net inversion.

$$C_6H_5\underset{\underset{(+)}{Me}}{CH}-Cl \xrightarrow{MeCO_2H} C_6H_5\underset{\underset{88\%\ racemisation}{12\%\ net\ inversion}}{\underset{Me}{CH}}-OCOMe$$

- Adding the much more powerfully nucleophilic $MeCO_2^-$ (as $MeCO_2^-Na^+$) to the reaction mixture was found to result in: (a) no increase in the overall reaction rate and (b) no increase in the proportion of net inversion. which suggest that the reaction will be SN1 mechanism not SN2.

- This strongly suggests, the inversion that is observed does not stem from part of the overall reaction proceeding via an S_N2 pathway simultaneously with the (major) S_N1 mode. If it did, we would expect the change to a much more powerful nucleophile ($MeCO_2H \rightarrow MeCO_2^-$) to lead to marked increases in both (a) and (b) above.

22-09

S_Ni → intramolecular nucleophilic substitution

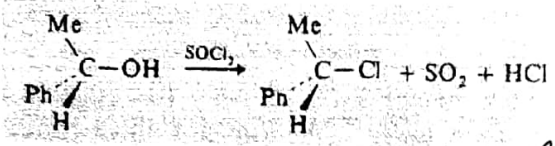
22-Oct-16

There are some reactions that proceed with actual retention of configuration, i.e. in which the starting material and product have the same configuration. This is known as S_Ni mechanism or Retention of Configuration.

TU-16

S_Ni Mechanism: Retention of Configuration

- A number of cases are known that proceed with actual retention of configuration, i.e., in which the starting material and product have the same configuration.
- This has been shown to occur is in the replacement of OH by Cl through the use of thionyl chloride, $SOCl_2$:



$$\text{Rate} = k_2[\text{ROH}][SOCl_2]$$

57

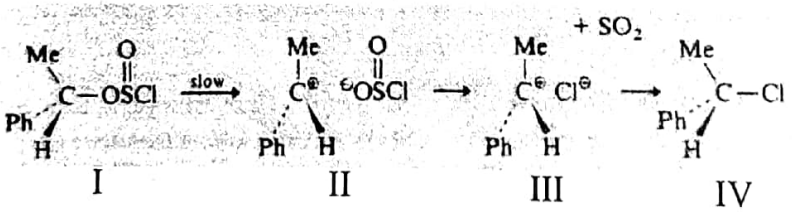
F-12

→ rate ∝ [ROH][$SOCl_2$]

* S_N2 में हमेशा totally inversion होगा।

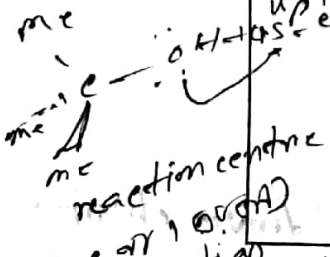
* S_N1 " " racemization & inversion दोनों।

- Clearly cannot proceed by the simple S_N2 mode lead to inversion of configuration in the product.



→ but mechanism S_N1 या S_N2 नहीं

- Carrying out the reaction under milder conditions allows of the isolation of an alkyl chlorosulphite, ROSOCl (I) and this can be shown to be a true intermediate.



$R^+ SO_2Cl$

58

$[O-S-Cl \rightarrow SO_2 + Cl]$

Finally Cl^- attack on the same side of R^+ from which SO_2Cl departed and resulting the retention of configuration.

- The rate at which the alkyl chlorosulphite intermediate (I) breaks down to the product, RCl (IV) is found to increase with increasing polarity of the solvent and also with increasing stability of the carbocation R^+: an ion pair, $R^+ {}^-OSOCl$ (II) is almost certainly involved.
- Finally attack by Cl^- is likely to occur on the same side of R^+ from which ^-OSOCl departed, i.e., with retention of configuration.
- It is interesting that if the $SOCl_2$ reaction on ROH is carried out in the presence of pyridine, the product RCl is found now to have undergone inversion of configuration.

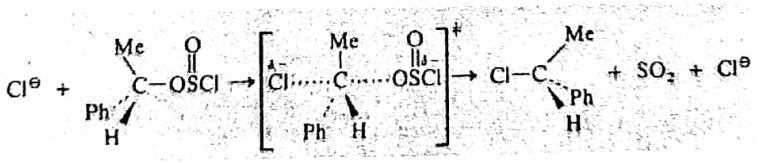
59

* কেন Pyridine থাকলে inversion হয়?

Pyridine এর উপস্থিতিতে normal S_N2 হয়।

alkyl chlorosulphite (Rosacel)

- This occurs because the HCl produced during the formation of (I) from ROH and $SOCl_2$ is converted by pyridine into $C_5H_5NH^+Cl^-$ and Cl^-, being an effective nucleophile, attack (I) 'from the back' in a normal S_N2 reaction with inversion of configuration.

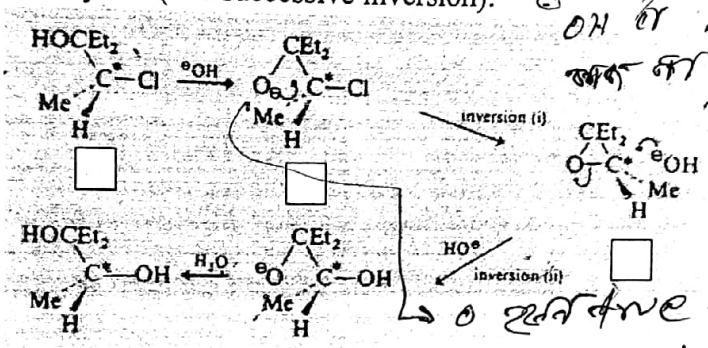


60

Pyridine এর presence এ নতুন pyridinium chloride তৈরি হয় এবং এটি [illegible] । then [illegible] ion এসে attack করে।

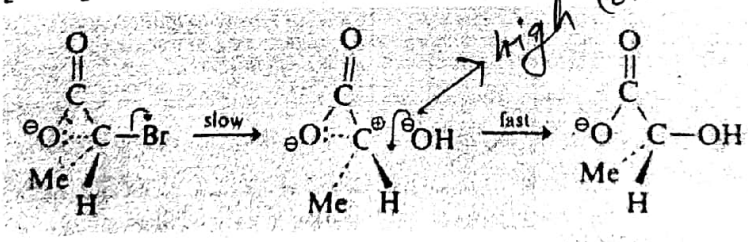
Neighbouring Group Participation: 'Retention'

- The neighbouring group can use its electron pair to interact with the 'backside' of the carbon atom undergoing substitution, thus preventing attack by the nucleophilic reagent, attack can thus take place only 'from the front side', leading to retention of configuration. Hydrolysis of chlorohydrin (two successive inversion):

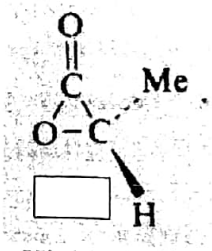


61

- Hydrolysis of 2-bromopropanoate anion at low [⁻OH]: rate is independent of [⁻OH].



- As the concentration of nucleophile, [⁻OH], is increased an increasing proportion of normal S_N2 'attack from the back', with inversion of configuration is observed.

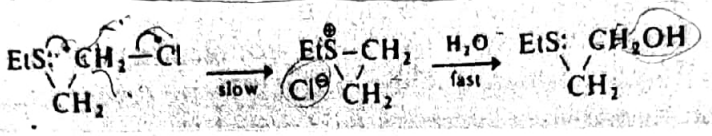


Highly labile α-lactone

62

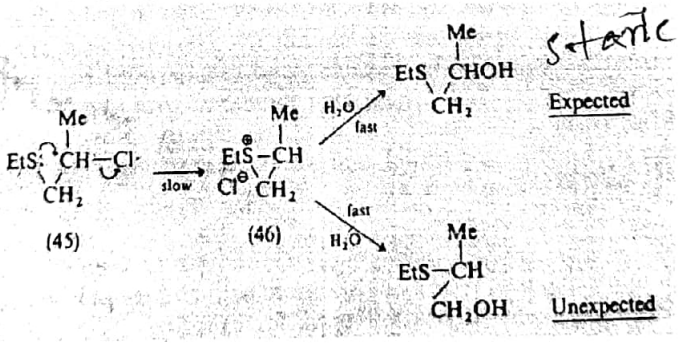
Internal nucleophile attack is very much favoured than the external nucleophile.

$EtSCH_2CH_2Cl$ is found to undergo hydrolysis 10^4 times faster than $EtOCH_2CH_2Cl$ under comparable conditions and this has been interpreted as involving S: as acting as a neighbouring group:



By contrast, O: is sufficiently electronegative not to donate an electron pair (unlike O^- in RO^- and RCO_2^-) and hydrolysis of $EtOCH_2CH_2Cl$ thus proceeds via ordinary S_N2 attack by an external nucleophile – which is likely to be very much slower than the internal nucleophilic attack.

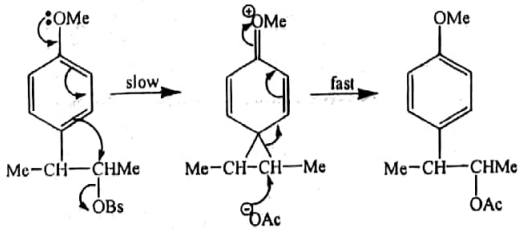
63



steric effect

maximum

The π orbital system of the benzene ring can also act as a neighbouring group.



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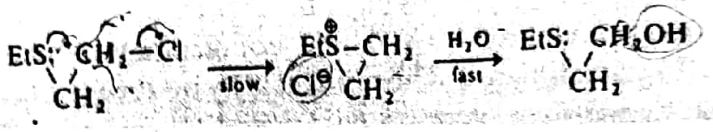

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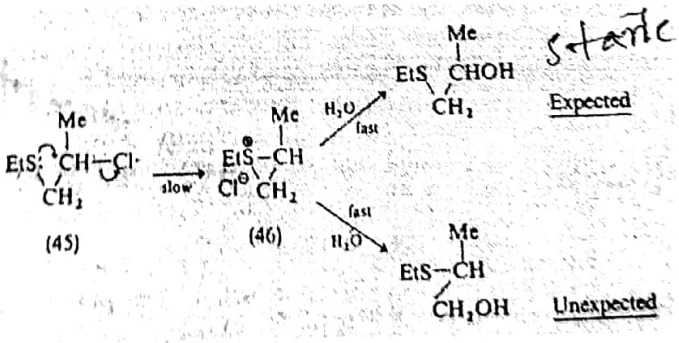
63



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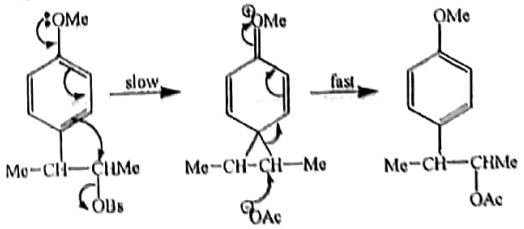

Basicity এর similarity & Desimilarity এর মধ্যে পার্থক্য?

Basicity এর similarity and desimilarity এর মধ্যে পার্থক্য?

Effect of Entering and Leaving Groups
The Entering Group

- In S_N2 displacement, the more strongly nucleophilic the reagent the more the reaction will be promoted.
- The nucleophilicity of a reagent might be expected to correlate with its basicity, as both involve the availability of electron pairs and the ease with which they are donated.
- Basicity involves electron pair donation to hydrogen whereas nucleophilicity involves electronpair donation to another atom, very often carbon; basicity involves an equilibrium (thermodynamic) whereas nucleophilicity usually involves a kinetic; basicity is likely to be little affected by steric influences whereas nucleophilicity may be markedly affected.

65

Similarity and dissimilarity of basicity and nucleophilicity?

Hard base : basic হিসেবে কাজ করে soft base: nucleophile হিসেবে কাজ করে

- Recently introduced the terms hard and soft bases: a hard base is one in which the donor atom is high of electronegativity, low polarizability and is hard to oxidize, i.e., ^-OH, ^-OR, R_3N:; while a soft base is one in which the donor atom is of low electronegativity, high polarizability and is ease to oxidize, i.e., RS^-, I^-, SCN^-; for a given degree of basicity, softness promotes nucleophilicity.
- If the attacking atom is the same then the two run reasonably in parallel and we find the stronger the base the more powerful the nucleophile.

$$EtO^{\ominus} > PhO^{\ominus} > MeCO_2^{\ominus} > NO_3^{\ominus}$$

66

Hard base = strong electron negative atom.

Soft base = nucleophile বেশি হবে।

- A shift in mechanistic type can also occur with change of nucleophile, thus a displacement that is S_N1 with, for example, H_2O:, HCO_3^-, $MeCO_2^-$, etc., may become S_N2 with ^-OH or EtO^-.
- Nucleophilicity is found to be very much affected by the size of the attacking atom in the nucleophile.

$$I^{\ominus} > Br^{\ominus} > Cl^{\ominus} \qquad RS^{\ominus} > RO^{\ominus}$$

- Size as well as electronegativity governs polarizability: as the atom increases in size the hold the nucleus has on the peripheral electrons decreases with the result that they become more readily polarizable, leading to the initiation of bonding at increasing nuclear separations.

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- Also the larger the nucleophilic ion or group the less its solvation energy, i.e., the more readily is it converted into the effective, largely non-solvated nucleophile.
- This combination of factors makes the large, highly polarizable, weakly solvated iodide ion, I^-, a very much better nucleophile than the small, difficulty polarizable, strongly solvated (H-bonding with a hydroxylic solvent) fluoride ion, F^-.
- Except the increase in reaction rate on transfer from a hydroxylic to a polar non-protic solvent to be much less for I^- than for example, Br^- or Cl^-: as is indeed found to be the case (Br^- is a better nucleophile than I^- in Me_2CO).

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A further interesting point arises with nucleophiles which have more than one suitable atoms through which they can attack the substrate.

$$[^{\ominus}X{=}Y \leftrightarrow X{=}Y^{\ominus}]$$

- In S_N1 reactions attack takes place on the carbocationic intermediate, R^+, through the atom in the nucleophile on which electron density is the higher.
- For example, halides that do not readily undergo S_N1 attack this can be promoted by use of the silver salt of the anion, e.g., AgCN, as Ag^+ promotes R^+ formation by precipitation of AgHal.

69

$$[^{\ominus}C{\equiv}\ddot{N} \leftrightarrow \ddot{C}{=}\ddot{N}^{\ominus}]$$

$$R{-}Br + Ag^{\oplus}[CN]^{\ominus} \xrightarrow[slow]{} AgBr\downarrow + R^{\oplus} + [CN]^{\ominus} \xrightarrow[fast]{} R{-}\overset{\oplus}{N}{\equiv}C^{\ominus}$$

- In the absence of such promotion by Ag^+, e.g., with $Na^+[CN]^-$, the resulting S_N2 reaction is found to proceed with preferential attack on the atom in the nucleophile which is the more polarizable.

$$NC^{\ominus} + R{-}Br \rightarrow \left[\overset{\delta-}{N}C\cdots R\cdots \overset{\delta-}{Br}\right]^{\ddagger} \rightarrow N{\equiv}C{-}R + Br^{\ominus}$$

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- This is understandable as, unlike S_N1, bond formation is now taking place in the T.S. for the rate-limiting step, for which ready polarizability of the bonding atom of the nucleophile is clearly important – the beginning of bonding at as great an internuclear separation as possible.
- Similarly, nitrite ion $[NO_2]^-$ is found to result in the formation of alkyl nitrites, R-O-N=O, under S_N1 conditions (O is the atom of higher electron density) and nitroalkanes, R-NO_2, under S_N2 conditions (N is the more readily polarizable).

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S_N1 : R-O-N=O
R-NO$_2$

The Leaving Group

- Changing the leaving group will clearly alter the rate of both S_N1 and S_N2 reactions, as breaking the bond to the leaving group is involved in the slow, rate-limiting step of both.
- The relative ability of Y as a leaving group, in R-Y, to be influenced by: (a) the strength of R-Y bond, (b) the polarizability of this bond, (c) the stability of Y^- and (d) the degree of stabilization, through solvation, in the T.S. for either S_N1 or S_N2.
- The observed reactivity (S_N2 or S_N1) sequence for halides R-I > R-Br > R-Cl > R-F suggests that here (a) and (b) above are probably more important than (c) and (d).

72

- The anions of strong ' oxygen acids' such as *p*-$MeC_6H_4SO_3^-$ (tosylate), $CF_3SO_3^-$ (triflate) are good leaving groups, with such O-leaving groups (c) and (d) above are of increased importance.
- Relative leaving group ability may vary with change of solvent, reflected the influence of (d).
- This variation in relative ability can be particularly marked on changing from a hydroxylic solvent to a bipolar, non-protic one (DMSO, DMF) as initial (c)/(d) domination may shift to (a)/(b) control.
- High polarizability makes I^- both a good entering and a good leaving group, it can thus often be used as a catalyst to promote an otherwise slow displacement reaction, e.g.,:

73

* stabilization of leaving group.
* o- less polarizable

weak nucleophile.
① O- of atom of H_2O is neutral molecule.

$$H_2O: + R{-}Cl \xrightarrow[slow]{} HO{-}R + H^{\oplus}Cl^{\ominus}$$

$$I^{\ominus} + R{-}Cl \xrightarrow{fast} I{-}R + Cl^{\ominus}$$

$$I{-}R + H_2O \xrightarrow{fast} H^{\oplus}I^{\ominus} + R{-}OH$$

- This is known as nucleophilic catalysis.
- Groups such as ^-OH, ^-OR, $^-NH2$ bonded to carbon by small, highly electronegative atoms of low polarizability (hard bases) can not normally be displaced directly by other nucleophiles.
- Thus ^-OH can not be displaced directly by Br^- but is displaced readily if protonated first.

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* less polarizable

① OH^-, ^-OR, $^-NR_2$ are not good leaving group. why

Know: How they good leaving group.?

- (a) Br^- is now attacking a positively charged species and (b) the very weakly basic H_2O is a much better leaving group than the strongly basic ^-OH.

$$Br^{\ominus} + R{-}\ddot{O}H \rightleftharpoons Br{-}R + {}^{\ominus}OH$$

$$H^{\oplus}\uparrow\downarrow \longrightarrow$$

$$Br^{\ominus} + R{-}\overset{\oplus}{\underset{\ddot{H}}{\ddot{O}}}H \longrightarrow Br{-}R + H_2O$$

- The well known use of HI to cleave ethers results from I^- being about the most nucleophilic species that can be generated in the strongly acid solution required for the initial protonation:

$$R{-}\ddot{O}Ph \overset{H^{\oplus}}{\rightleftharpoons} R{-}\overset{\oplus}{\underset{\ddot{H}}{\ddot{O}}}Ph \xrightarrow{I^{\ominus}} RI + PhOH$$

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Other Nucleophilic Displacements

- Attack by non-charged nucleophiles:

$$Me_3N: + Et{-}Br \longrightarrow Me_3\overset{\oplus}{N}Et + Br^{\ominus}$$

$$Et_2S: + Me{-}Br \longrightarrow Et_2\overset{\oplus}{S}Me + Br^{\ominus}$$

- Nucleophilic anions on positively charged species:

$$I^{\ominus} + C_6H_{13}{-}\overset{\oplus}{\underset{H}{O}}H \longrightarrow C_6H_{13}{-}I + H_2O:$$

$$Br^{\ominus} + Me{-}\overset{\oplus}{N}Me_3 \longrightarrow Me{-}Br + :NMe_3$$

$$H_2O: + PhN_2^{\oplus} \longrightarrow PhOH + N_2 + H^{\oplus}$$

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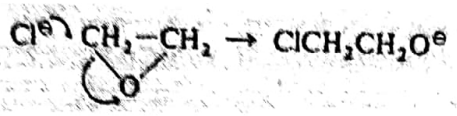
- Good leaving group other than halide:

$$MeCO_2^{\ominus} + ROSO_2C_6H_4Me\text{-}p \rightarrow MeCO_2R + p\text{-}MeC_6H_4SO_3^{\ominus}$$

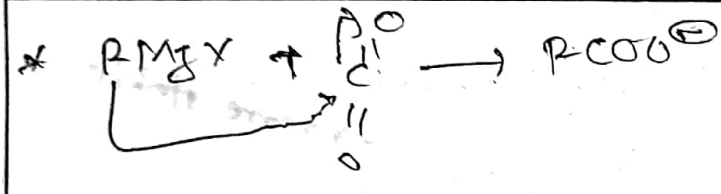
- Internal leaving group:



$$Cl^{\ominus}\ CH_2-CH_2 \rightarrow ClCH_2CH_2O^{\ominus}$$

- There are also nucleophilic displacement reactions of considerable synthetic importance in which the attacking atom in the nucleophile is carbon in either a carbanion, new carbon-carbon bonds are thus formed.

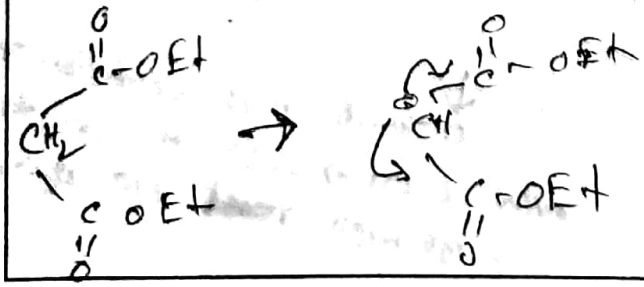
77



$$HC\equiv CH \overset{^{\ominus}NH_2}{\rightleftarrows} HC\equiv C^{\ominus} + Pr-Br \rightarrow HC\equiv C-Pr + Br^{\ominus}$$

$$CH_2(CO_2Et)_2 \overset{EtO^{\ominus}}{\rightleftarrows} (EtO_2C)_2CH^{\ominus} + PhCH_2-Br \rightarrow (EtO_2C)_2CH-CH_2Ph + Br^{\ominus}$$

$$\overset{\delta+\ \delta-}{BrMgPh} + C_6H_{13}-Br \rightarrow MgBr_2 + Ph-C_6H_{13}$$



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Nucleophilic Attack on Aromatic Species. Substitution of Hydrogen

- It is expected that attack by nucleophiles on an unsubstituted benzene nucleus will be much more difficult than by electrophiles.
- This is so (a) because the π electron cloud of the nucleus is likely to repel an approaching nucleophile and (b) because its π orbital system is much less capable of delocalizing (and so stabilizing) the two extra electrons in the negatively charged than the positively charged intermediate.

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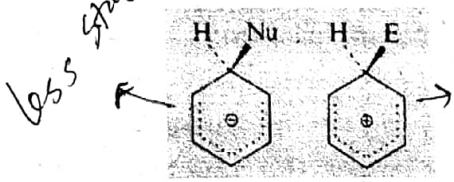
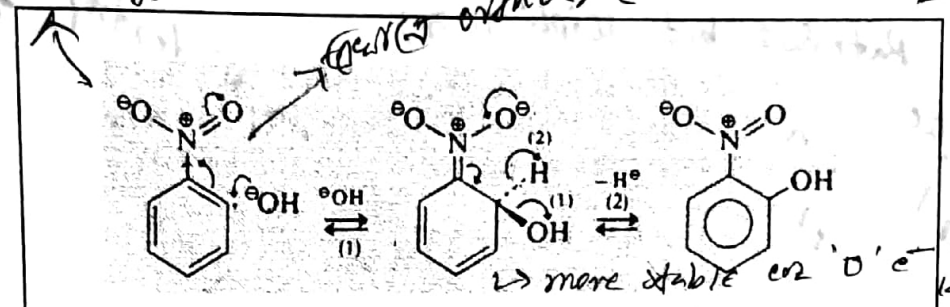


- Both (a) and (b) would be overcome to some extent if a sufficiently powerful electron-withdrawing substituent was present and nucleophilic attack might then become possible.
- It is found in practice that nitrobenzene can be fused with KOH in the presence of air to yield *o*- (plus a little *p*-) nitrophenol.

80

→ why we use air in this reaction?

22-Oct-16



- Other canonical states can be written but by far the most significant one is that shown above in which the (-) charge is accommodated by an oxygen atom of the nitro group.
- This can occur only if the attacking ^-OH enters the positions *o*- and *p*- to the NO_2 group.
- The intermediate can regain the aromatic condition by either ^-OH or H^- acting as a leaving group.

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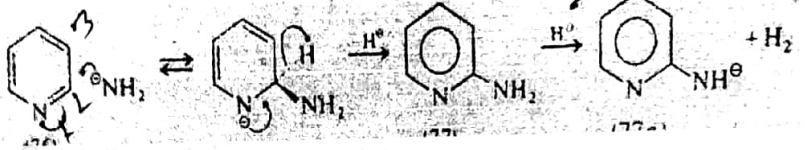
- H^- is a poor leaving group so the equilibrium tends to lie over to the left - ^-OH, being a better leaving group, is lost a lot more often than H^- - unless an oxidizing agent, e.g., air, KNO_3 or $K_3Fe(CN)_6$, is present to encourage the elimination of hydride ion and to destroy it as formed.
- Nitrobenzene can act as its own oxidizing agent but the yield of nitrophenol is then very poor.
- Pyridine is itself attacked by powerful nucleophiles, e.g., by $^-NH_2$ (sodamide, $NaNH_2$) in N,N-dimethylaniline as solvent – Tschitschibabin reaction.

82



- The leaving group, H^-, subsequently removes a proton from the introduced NH_2 group, thereby evolving H_2 and converting the amino group into its anion.
- This anion may ultimately be converted on treatment with H_2O into the desired 2-aminopyridine.

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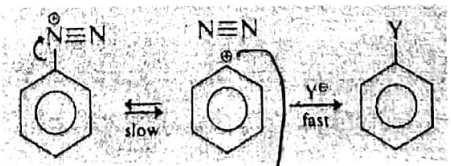
Substitution of Atoms Other Than Hydrogen

- One very common example is the displacement of N_2 in the reactions of diazonium salts, ArN_2^+, a very useful preparative series:

$$ArN_2^{\oplus} + Y^{\ominus} \rightarrow ArY + N_2$$

$$\text{Rate} = k[ArN_2^{\oplus}]$$

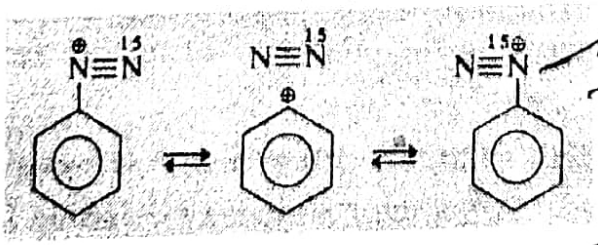
- The rate is independent of $[Y^-]$ and analogies to S_N1.



84

- Added nucleophiles, Cl^-, MeOH, etc., are found to affect the product composition but not the rate of reaction.
- The formation of the highly unstable phenyl cation [the (+) charge can not be delocalized by the π orbital system] is somewhat surprising but the driving force is provided by the extreme effectiveness of N_2 as a leaving group. It is significant that this appears to be the only reaction by which simple aryl cations can be generated in solution.
- The aryl cations are highly reactive and thus unselective towards nucleophiles: thus the selectivity between Cl^- and H_2O (k_{Cl^-}/k_{H_2O}) is only 3 for $C_6H_5^+$ compared with 180 for Me_3C^+.

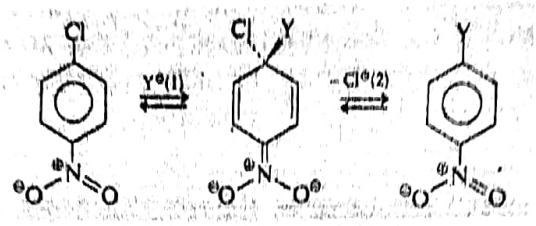
- The very high reactivity of $C_6H_5^+$ is reflected in its ability to recombine with N_2, i.e., the decomposition of the diazonium cation is reversible.



- The step is reversible, was demonstrated by observing the partial scrambling of the ^{15}N label.
- A particularly useful displacement reaction on ArN_2^+ is the introduction of F into the benzene nucleus (not possible by direct reaction with F_2, vigorous reaction).

$$ArN_2^{\oplus}BF_4^{\ominus} \xrightarrow{\Delta} Ar{-}F + N_2\uparrow + BF_3\uparrow$$

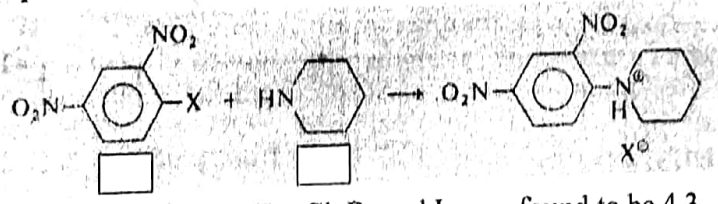
- The most common aromatic nucleophilic displacement reactions involve the displacement of Hal^- from a halide activated by electron-withdrawing groups.



$$Rate = k[ArX][Y^{\ominus}]$$

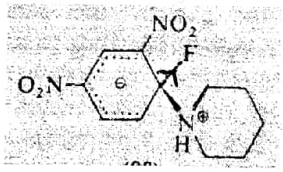
S_N2 (aromatic)

- Direct support for a stepwise pathway is provided by comparison of the rates of reaction of a series of substrates having different leaving groups with same nucleophile, e.g., 2,4-dinitrohalogenobenzene with piperidine.

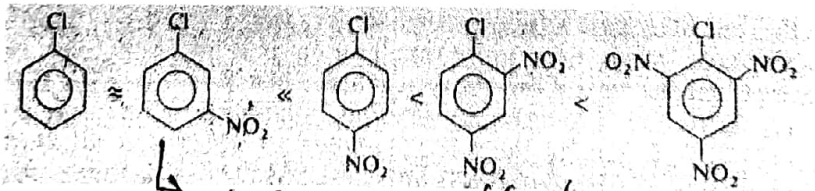


- The relative rates for X = Cl, Br and I were found to be 4.3, 4.3 and 1.0, respectively; breaking of the C-X bond thus cannot be involved in the rate limiting step of the reaction, or we should expect significantly bigger rate differences and in the sequence I > Br > Cl.

- The reaction, in this case, can not be one-step and in the two-step pathway suggested above, step (1) - attack by the nucleophile – would have to be rate-limiting.
- It is interesting too, to observe that the rate of the above reaction when X = F is 3300.
- This results from the very powerfully electron withdrawing F speeding up step (1): (a) by making the nuclear carbon to which it is attached more positive and hence more readily attacked by a nucleophile and (b) by helping to stabilize the anionic intermediate.

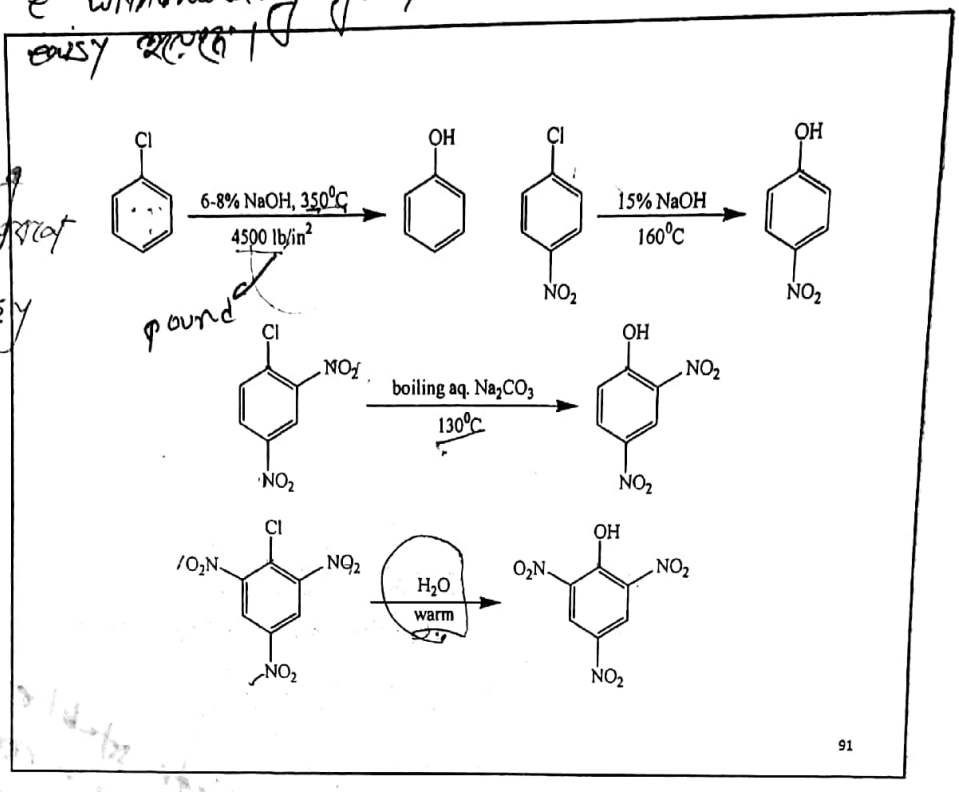


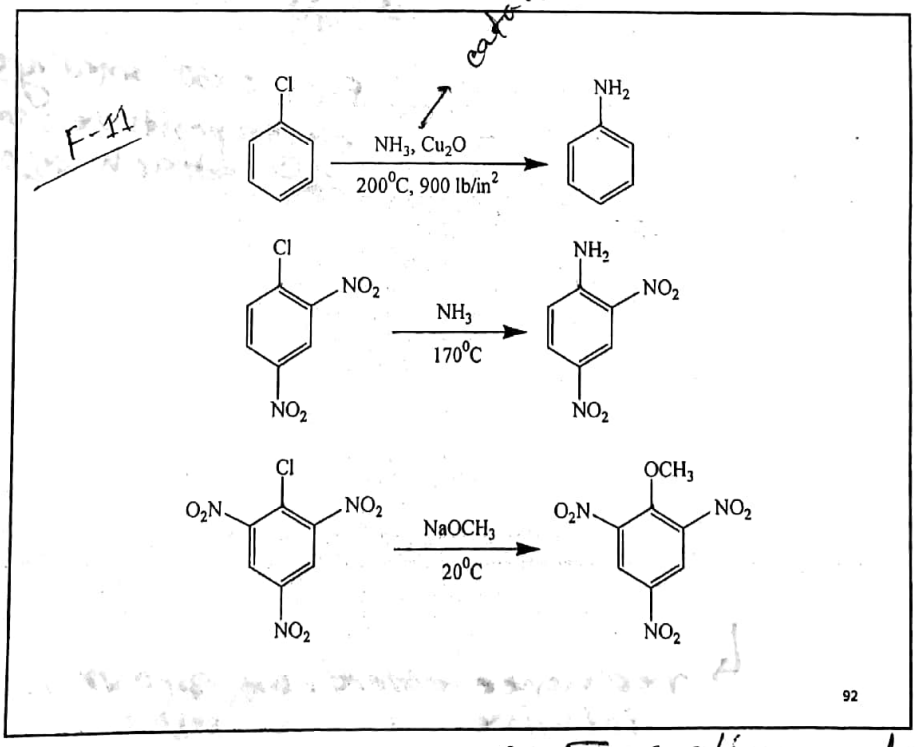
- Nucleophilic aromatic substitution is in fact an addition/elimination process very similar to electrophilic aromatic substitution except for the different attacking species.
- The sequence for nucleophilc aromatic substitution:



e⁻ withdrawing group থাকলে reaction easy হয়ে।

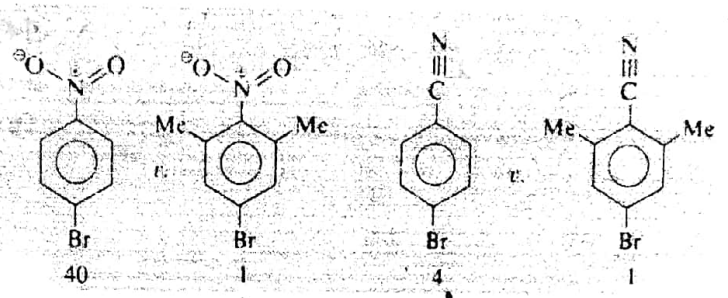
"e⁻withdrawing group থাকায় reaction easy হয়েছে।"





* e⁻ with drawing group থাকলে reaction rate কমে- why?

- Mesomeric interaction with an electron-withdrawing substituent will be reduced or inhibited if the p orbital on the atom adjacent to the nucleus, e.g., N in NO_2, is prevented from becoming parallel to the p orbital on the nuclear carbon to which it is attached.
- Thus the following relative rates of nucleophilic attack are observed:



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- The rate difference between last two compounds is very small as the Me groups do not prevent mesomeric electron-withdrawal by the linear CN group.
- The rate difference is much more pronounced between first two compounds as the Me groups prevent the oxygen atoms of the nitro group lying in the same plane as the nucleus, p overlap between N and the adjacent C is thus markedly reduced.
- Rate differences of as much as 10^9 have been observed on changing the solvent from MeOH to Me_2SO. Chlorobenzene will thus react readily under these conditions with Me_3CO^-.

$$Me_3CO^{\ominus} + Ph{-}Cl \xrightarrow{DMSO} Ph{-}OCMe_3 + Cl^{\ominus}$$

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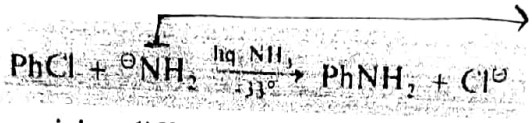
Substitution via Aryne Intermediates

- The relative inertness of unactivated aromatic halide towards nucleophiles, under normal conditions, is in sharp contrast to their marked reactivity towards nucleophiles that are also very strong bases.
- Thus chlorobenzene is readily converted into aniline by reaction with $^{-}NH_2$ ($NaNH_2$) in liquid ammonia at -33°:



- This surprising difference in reactivity suggests the possibility of a reaction pathway other than S_N2 (aromatic).

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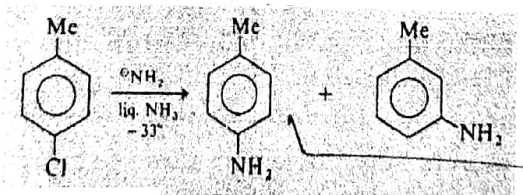
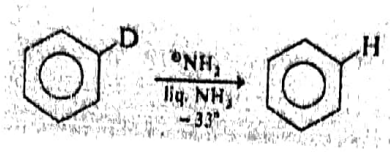
- *p*-Chloromethylbenzene undergoes the same reaction to give not only the expected *p*-aminomethylbenzene but also unexpected *m*-aminomethylbenzene.

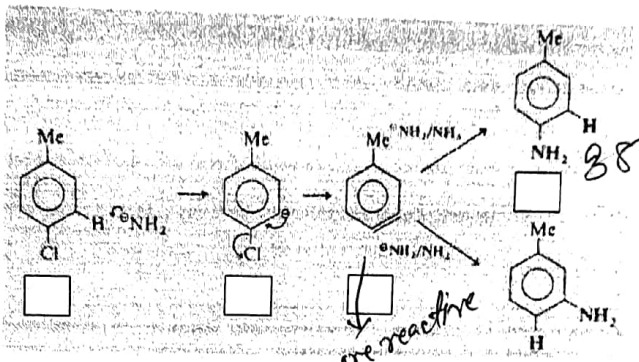


- Two products are found not to be interconvertible under the conditions of the reaction.
- $^{-}NH_2$ is known to be able to remove protons (deuterons) from a benzene ring.

96



- Suggests that attack by $^{-}NH_2$ may be as a base on H *ortho* to Cl, rather than as a nucleophile on C of the C-Cl bond.



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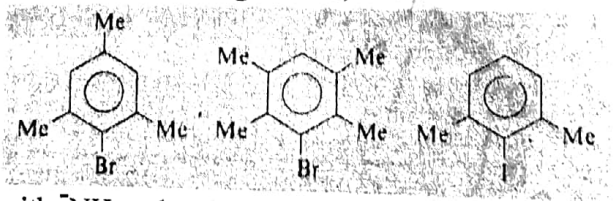
Cl-সাপেক্ষে ortho এবং meta- ✓ Aryne intermediate
position এ -NH2 attack হয়ে, meta position are more suitable.
যে জন্য 2টি product পাওয়া যায়, so unexpected product are major.

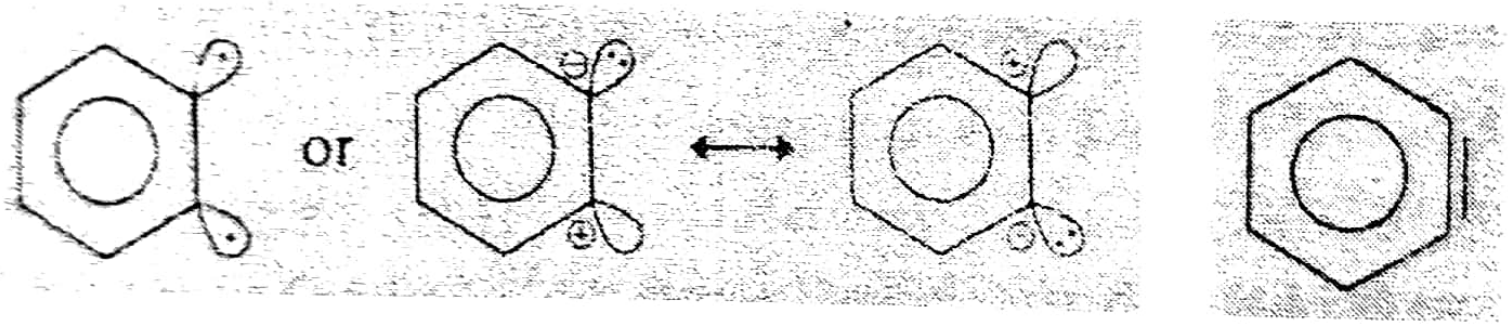
- We should not expect the relative proportions of the two alternative products to be the same because this aryne is not a symmetrical intermediate, i.e., the two possible positions of attack by $^{-}NH_2$ are not identical.
- This is clearly an elimination/addition mechanism in contrast to the addition/elimination of S_N2 (aromatic).
- Direct evidence in support of the aryne pathway is provided by the fact that the following halides,



react with $^{-}NH_2$ only under conditions much more vigorous than can be explained by any steric effect exerted by their *o*-Me groups.

98

- None possesses an H atom *ortho* to the halogen: a requirement essential for initiating reaction via an aryne intermediate, as we saw above.

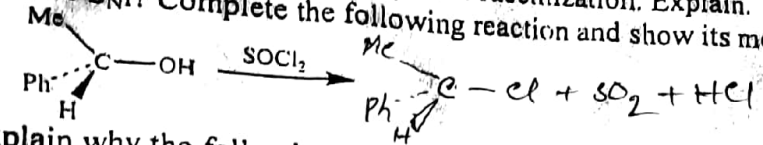


- sp^2-sp^2 side wise overlap: very poor → form very weak bond.
- Arynes are unstable and highly reactive.
- Stability of benzene ring, i.e., π-delocalization preserved.

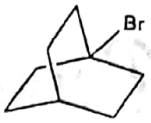
Course No.: 330, Full Marks: 20, Time: 60 minutes, Date: 22 [illegible], 2017

2nd Tutorial Examination - 2016

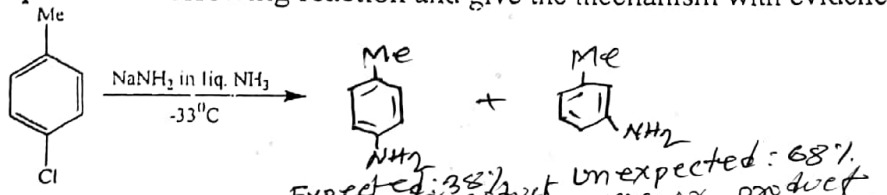
1. Give the reaction mechanism of the hydrolysis of *t*-butyl bromide in presence of base.
2. Solvolysis of (+)-PhCHMeCl leads to 98% racemization in aqueous acetone while (+)-C_6H_{11}CHMeCl leads to only 34% racemization. Explain. 3
3. What is S_Ni? Complete the following reaction and show its mechanism. 3



4. Explain why the following compound is not hydrolyzed even by a hot and aqueous solution of KOH. 2



5. Which reagent in each of the following pairs is more nucleophilic and why? 2
   (i) $(CH_3)_3B$ and $(CH_3)_3N$ (ii) H_2O and H_2S.
6. Allyl halides are more reactive than n-propyl halides in both S_N1 and S_N2 reactions - explain. 3
7. The π-orbital system of the benzene ring can act as a neighbouring group in nucleophilic substitution. Explain with suitable example. 2
8. Complete the following reaction and give the mechanism with evidence. 3



Fi-2016|15

(a) What is the salient difference between S_N1 and S_N2 pathways? 3
(b) How do you make a kinetic distinction between the operation of the S_N1 and S_N2 modes? 1
(c) The rate of solovolysis of the tertiary halide, Me_3CBr, is found to be 3×10^4 times faster in 50% aqueous ethanol than in ethanol-explain.
(d) For the S_N2 mode, increasing solvent polarity is found to result in slight decrease in reaction rate-why?
(e) Why does polar non hydroxylic DMF have marked effect on, (increased 4.5×10^4 fold) S_N2 reaction of MeI with N_3^- at 0° C when compared the reaction with MeOH?
(f) Why does S_N2 reaction rate decrease as the following series is traversed?

CH_3-Br $MeCH_2$-Br Me_2CH-Br Me_3C-Br

2015

6. (a) The aikaline hydrolysis of 1,2-chlorohydrin ($MeCHClCEt_2OH$) is found to yield the 1,2-diol. $MeCH(OH)CEt_2OH$ with retention of configuration-explain.
   (b) An overall ANTI hydroxylation takes place during the reaction of trans-2-butene with a peroxyacid-explain.
   (c) $EtSCH_2CH_2Cl$ is found to undergo 10^4 times faster than $EtOCH_2CH_2Cl$ under comparable condition-explain.
   (d) In S_N2 reactions of alkyl halides RX, the order of reactivity is $CH_3X>1^0>2^0>3^0$ - explain.
   (e) [illegible] reaction follows racemization with partial inversion of configuration.

Give proper explanation

(a) The following bicyclic halogenoalkane is almost unreactive to nucleophilic substitution reaction.

5. (a) Show the mechanism of bromination of 2-butene; give evidences in favour of your proposed mechanism. 2+1

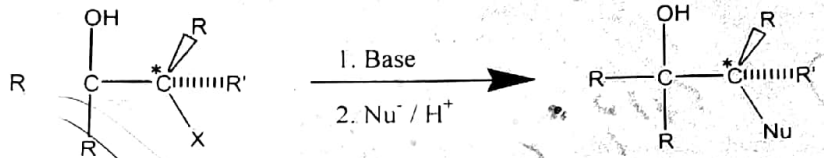
(b) Explain how the detection of intermediates help in determining the mechanism of a reaction. 2

(c) Draw a schematic representation of a S_N2 mechanism reaction. 2

(d) What effects will you observe on changing solvent from non-polr to polar solvent for a S_N2 reaction. 2

(e) Is there any effect of leaving group in a substitution reaction? Explain with example. 3

(f) Explain the formation of the following product that shows the retention of configuration.



(Retention of configuration)

4. (a) Differentiate between S_N1 and S_N2 reactions in relation to mechanism and stereochemistry. 4

(b) Explain the fact that the rate of substitution of the reaction of CH_3I with N_3^- is increased by a factor of 10^5 when the solvent is changed from CH_3OH to DMF even at 0°C. 4

(c) Formulate the product formation of the following reaction: 2

$CH_3CH{=}CHCH_2OH + SOCl_2 \rightarrow ?$

(d) Solvolysis of (+)-PhCHMeCl leads to 98% recimisation in aqueous acetone while (+)-C_6H_{13}CHMeCl leads to only 34% racemisation – explain. 2

(e) Explain the fact that hydrolysis of 2-bromopropionate anion at low [OH^-] proceeds with retention of configuration. 2

5. (a) Discuss the S_N2 reaction mechanism with special reference to 2+2

(i) structure of the substrate

(ii) stereochemistry

(b) In what respect S_N2 reactions differ from S_N1 reactions? 3

(c) Draw the potential energy diagram for both S_N1 and S_N2 reaction. 2

(d) Show the reaction mechanism of

What will happen if the above reaction is carried out in presence of pyridine? 3+2

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